

Automatic Detection of Tea Leaf Diseases using Deep Convolution Neural Network

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Abstract - In India, tea is the most consumed and popular beverage, all over the world too. It is one of the popular and widely cultivated perennial plantation crop in our state. Their productions are heavily affected and destroyed by different diseases. Deep learning is a machine learning technique that teaches a computer with the help of pre-existing data and enables the system to do what comes naturally to humans. The main challenge faced in deep learning is that it needs a large amount of training data. In this model, we are going to propose a unique idea to detect and classify diseases in tea leaves by incorporating deep learning techniques. The critical processes in this tea leaf disease classification are health monitoring and disease detection and are essential for sustainable agriculture. Manually observing the tea leaf diseases is a tedious and time taking process that requires skilled workers, extra time, and manpower with knowledge about tea leaf diseases. Hence, image processing models were widely utilized for these kinds of disease detections. In this proposed work, we have applied the Convolutional Neural Network (CNN) model with 1 input layer, 4 convolution layers, and 2 fully connected layers. The image is passed to the input layer. The convolution layers mainly extract features from the input image in the dataset and the output layer classifies the given image to 8 classes such as the normal leaf, Algal leaf spot, Gray blight, White spot, Brown blight, Red scab, Bud blight, and Grey blight.

Keywords: Deep learning, convolutional neural network, CNN, tea leaf, disease identification

I. INTRODUCTION

The traditional machine learning (ML) algorithms have been applied for various fields [5,6,14]. The development of several advanced image processing techniques introduced a new era for the usage of deep learning in agricultural fields,

especially in plant/leaf disease identification.

To reduce food demand and to get high productivity, monitoring of the agricultural problems is an essential one and it is possible by new technologies [12]. Nowadays, artificial intelligence techniques like machine learning and various deep learning algorithms have been mainly used in a wide variety of agricultural operations [13]. Among these techniques, deep learning achieves better developments in the agricultural field of research because of its automatic feature extraction capability and also consists of several deep learning algorithms.

Crop plant disease detection in the prior stage is one of the most important roles in agriculture. Crop affected by diseases results in a great loss of economic growth [11]. Therefore, quick and accurate detection diagnosis of the disease prevents the quality and quantity of the product. The visual view of the affected leaf is difficult and may fail to find it and cure the disease successfully and disease detection in the laboratory takes a process for disease culturing which may fail to provide results within the required time.

Tea is popularly known as "chai," is one of the major crops which is cultivated only in certain climatic conditions and regions. It is grown in almost every hill station around the country than other crops. Tea diseases have a major effect on tea production and exporting. The diseases can be detected in many parts of crops like leaf, stem. Observing the disease with normal eye and detecting the disease results in the inaccurate classification of the disease. This leads to the wrong usage of pesticides, fertilizer. Thus, the identification of tea leaf diseases plays an important role in the production of high quality and high yield of tea.

Many machine learning algorithms have been used for tea leaf disease classification. Compared to the machine learning

models, deep learning architectures including AlexNet [7], Visual Geometry Group (VGG)[8], Inception-v4[10], and ResNet[9] got good results for the image based classification problem. In this paper, we have applied CNN to classify the 7 types of disease in tea leaf using CNN is proposed along with normal leaf identification. We have considered diseases like Gray blight, White spot, Brown blight, Bud blight, Leaf blight, and Red scab using convolution neural network (CNN).

II. LITERATURE REVIEW

In [1], the authors proposed a model for plant disease detection and classification with the help of healthy leaf images and leaves with diseases. They have taken 25 different types of plants with 87,848 images which include plants diseases, and also healthy plants. Their model performance has been reached up to 99.53% accuracy. Due to the high level of performance, CNN is highly recommended for the detection of plant diseases.

In [2], the authors have done a review on the usage of new techniques in imaging processing and computer vision approaches for plant diseases classification. In [3], Chen et al. proposed GMDH-Logistic model, which is a self-adaptive classification method used for plant disease detection. In this, the authors introduced a model approach for the automatic detection, recognition, and classification of plant leaf diseases by image process method we perform the feature engineering analysis and build the system for the prediction models.

Plant disease classification through leaf symptoms plays an important role. In [4], Zhang et al., have introduced a novel segmentation method of hybrid clustering to divide the given color image into several pixels to improve the classification accuracy and also to reduce the different segmentation carried out by EM algorithm.

III. PROPOSED WORK

CONVOLUTION LAYER

The convolution layer contains numerous filters used to extract the significant features from the given input images. The height and width of the filters are smaller than the input volume of the image. Each filter is manipulated with the input image to determine the feature map made of neurons.

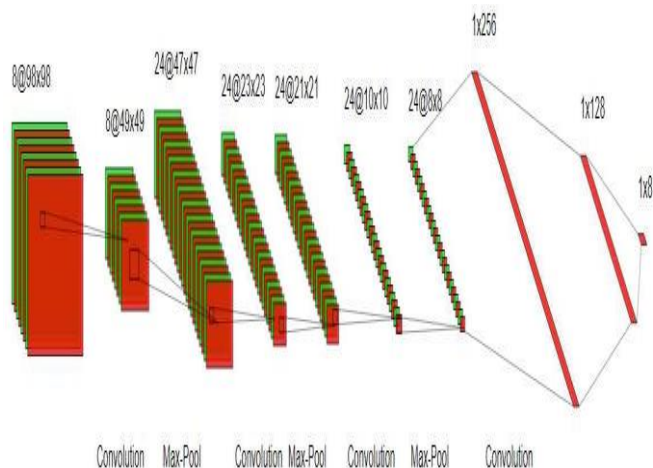


Figure 1. Proposed CNN architecture

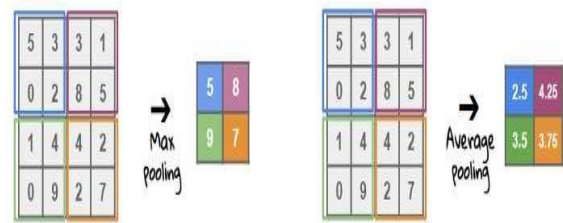


Figure 2 Pooling

POOLING LAYER

Pooling is the process of merging that helps to reduce the data. If we take the maximum value from the sub-matrix, then it is known as max pooling. If we take the average values, then it is average pooling. This is clearly shown in Figure 1.2

FLATTENING LAYER

Flatten layer converts the 2D array data into a 1-dimensional vector which is given as input for the fully connected feed-forward neural network. Flatten the output of the convolutional layers creates a single long feature vector. Using this data, the neural network is trained to identify the different tea leaf diseases.

RECTIFIED LINEAR UNIT(ReLU)

ReLU is one of the popular activation functions used in the CNN network. Unlike other activation functions, it does not activate all the neurons at the same time. It means that if the input to the ReLU function is negative, then it will convert it to zero and the neuron does not get activated.

FLOW DIAGRAM

Figure 3 represents the flow diagram of the proposed system for the detection of tea leaf diseases. It consists of different phases namely pre-processing, feature extractions, prediction, and plotting pathogen area for detecting disease. The CNN architecture is a layered approach in which the layers are processed sequentially.

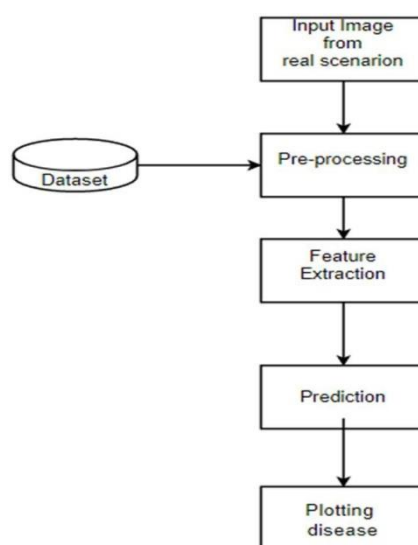


Figure3. Flow diagram for disease classification Pre-Processing

The images in the dataset are not standardized because it comes from a different source. So all the images need to be standardized and fed into the model. To standardize the images the simple pre-processing technique is used. There are different pre-processing techniques like resizing the images, scaling the images, normalizing the image, and dimensionality reduction. The larger image will occupy more space, and thus increases the time complexity and resulting in a large neural network. So, resizing of images is used to standardize the images. In the proposed system, all the images are resized as 100 X 100 X 3. The pre-processed images are next given as input to the convolution layer for the feature extraction process.

Feature Extraction

The feature of the images can be extracted using Convolutional Neural Network (CNN) architecture. The input image is typically given in a 2D array of neurons which denotes a set of pixels of the image. After preprocessing, the features of the images are extracted using CNN architecture which recognizes and classifies the images. CNN architecture has different layers namely convolution, pooling, and fully connected layer to do the feature extraction. Among these, the convolution layer has different filters to detect the particular feature at every location on the input image. The number of filters used in the convolution layers is mostly 64 or 128 which increment exponentially the number of filters for the layer. The filters divide the input as 2 X 2 with the default stride 1X 1. Each convolution layer embeds an activation function and pooling layer.

Among these, the activation function transforms the weighted sum of input to its higher or lower value like 0 or 1. There are three types of activation function namely binary step function, linear activation function, and non-linear activation function. Among these, non-linear activation functions are used to create multiple layers of neurons to process complex datasets such as images, videos, and audios in order to give high accuracy. Non-linear activation function had different functions namely sigmoid, hyperbolic tangent, ReLU, leaky ReLU, parametric ReLU, softmax, and Swish. Among these, ReLU function is used for all convolution layers because of its computational efficiency than any other activation functions. The output of the activation function is given to the pooling layer helps to reduce only the spatial information, and hence there is only less chance of over-fitting. Moreover, it is a downsampling process that reduces the size of data.

The output from the last convolution layer is fed as the input to the flattening layer. The flattening layer converts the 2D matrix into a vector format. It must be done before entering into the fully connected layer. The first fully connected layer consists of 2048 nodes activated by ReLU activation function. The second layer acts as the output layer. The activation function used in the output layer is softmax function and the output from the softmax function is a probability ranges between 0 to 1. The label which has the highest probability is assigned as the class of input image. The output layer is defined by several classes to be classified where the number of classes is 8 which includes normal healthy leaves and leaves with different diseases such as Algal leaf spot, Gray blight, White spot, Brown blight, Red scab, Bud blight, Grey blight.

IV. DATASET DESCRIPTION

The dataset is taken from the CIFAR-10. It contains all plant disease leaves from that tea leaf disease images are taken. Some classes of images are captured using a camera directly from the field. To create more number of images, we have applied data augmentation techniques. We have a total number of 860 images in the dataset. Among that, we have used, 80% of the images (688 images) for training, 20% of the images (172 images) for testing. The number of images in each class is given below: Figure 5 to 12 shows the tea leaves with different types of diseases.

Table 4Dataset

Class	Disease	No of images
1	Healthy leaf	115
2	Algae leaf spot	110
3	Gray Blight	90
4	White Spot	93
5	Brown Blight	121
6	Red Scab	97
7	Bud Blight	113
8	Leaf blight	121

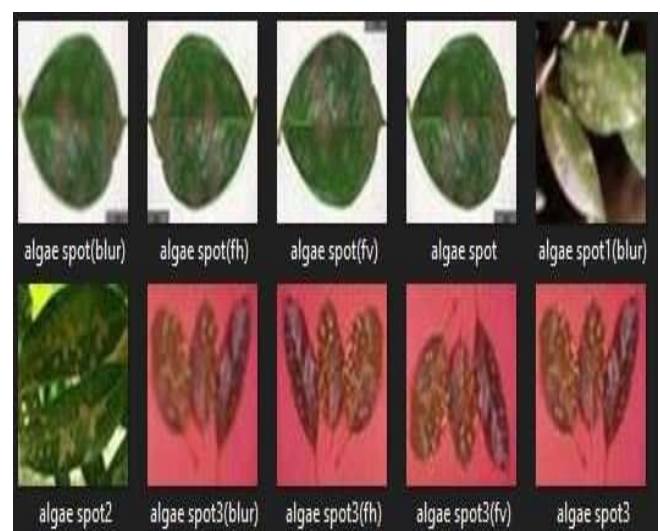


Figure 4 Algae leaf spot (sample images)

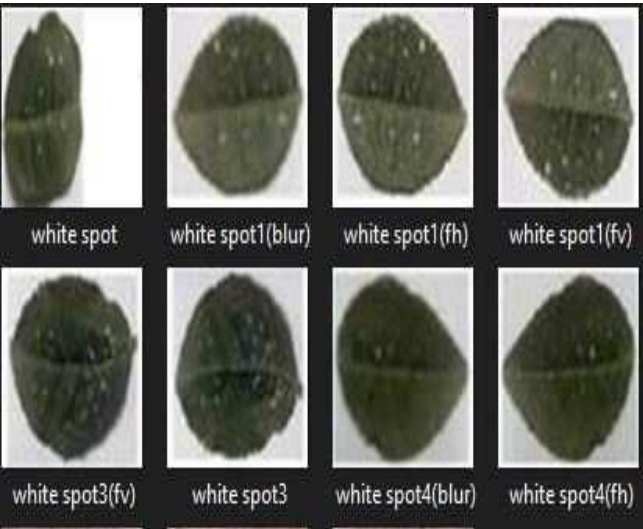


Figure 5 White spot(sample images)



Figure 8 Grey blight(sample images)



Figure 6 Leaf blight(sample images)



Figure 9 Bud blight(sample images)

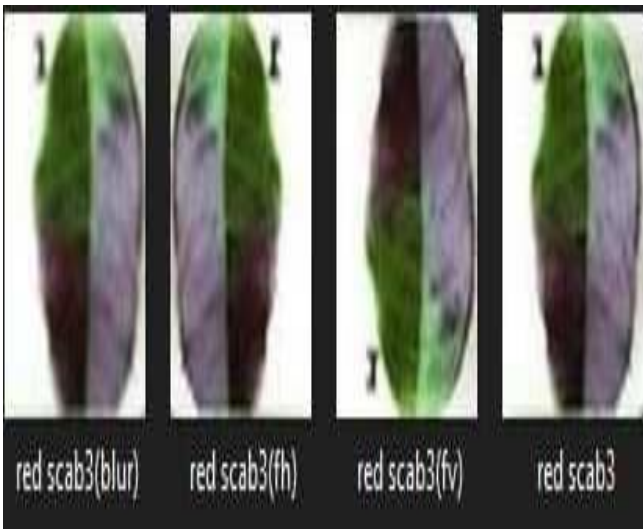


Figure 7 Red scab(sample images)

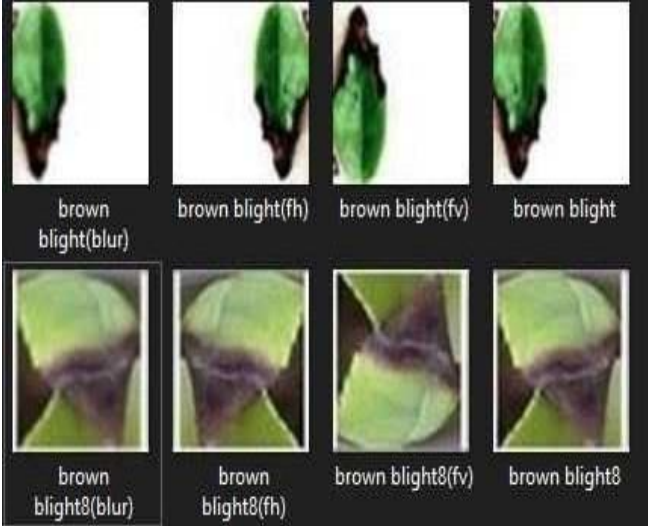


Figure 10 Brown blight (sample images)

V. PERFORMANCE ANALYSIS

Accuracy is treated as one of the most important parameters for analyzing the proposed work. The accuracy is the ratio between correctly identified image classes (sum of true positive and true negative) to the total number of images considered.

$$\text{Accuracy} = \frac{\text{TruePositive}(TP) + \text{TrueNegative}(TN)}{\text{TotalNumberOfImages}} \times 100$$

True Positive (TP): The tea leaves with diseases are identified correctly as affected with the correct label.

True Negative (TN): The healthy tea leaves are identified correctly as not affected.

False Positive (FP): The healthy tea leaves are misclassified as affected one.

False Negative (FN): The tea leaves with diseases are misclassified as normal healthy leaves.

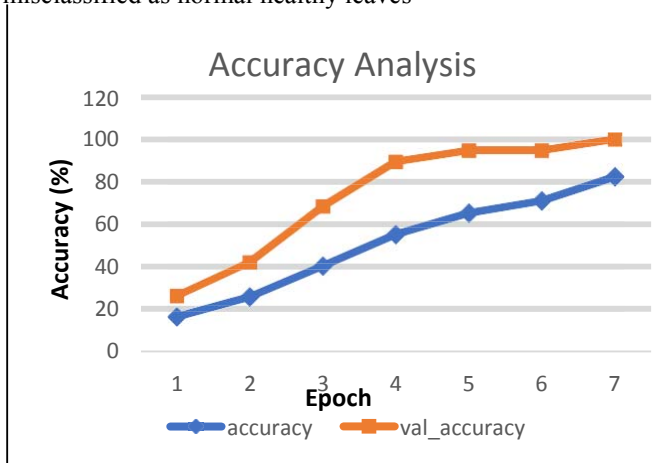


Figure 11 Accuracy Analysis

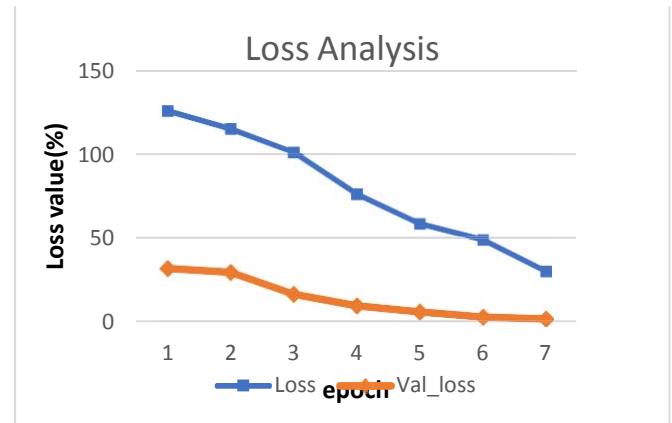


Figure 12 Loss Analysis

Figure 13 and 14 shows the performance of the proposed model in terms of accuracy and loss against epoch. An epoch is one complete pass of entire training of the data. It is very clear that the accuracy of the proposed work is improved when the number of epochs is increased. Usually, the training accuracy continues to improve because the model to find the best fit for the training data tends to overfit. Epoch value should be chosen in such a way that the model should not overfit.

VI. CONCLUSION

The ultimate goal of the proposed work is to classify and improve the accuracy of correctly predicting the diseases. The proposed model considers several tea leaf diseases and achieves the correct identification with 94.45% accuracy. This can be achieved with deep learning CNN architecture. The performance of the proposed is further improved by varying the number of layers, the learning rate parameter, and the optimizer used. Further, the results of the model can be compared with the existing CNN architecture by applying the transfer learning concept.

The proposed model with accuracy and loss for 8 epochs is given below:

Epoch 1/8

15/15 [=====] - 9s 112ms/step - loss: 125.7684 - accuracy: 0.1645 - val_loss: 31.9875 - val_accuracy: 0.2632

Epoch 2/8

15/15 [=====] - 1s 52ms/step - loss: 115.1528 - accuracy: 0.2585 - val_loss: 29.5600 - val_accuracy: 0.4211

Epoch 3/8

15/15 [=====] - 1s 52ms/step - loss: 101.1699 - accuracy: 0.4030 - val_loss: 16.5117 - val_accuracy: 0.6842

Epoch 4/8

15/15 [=====] - 1s 51ms/step - loss: 76.1560 - accuracy: 0.5507 - val_loss: 9.6421 - val_accuracy: 0.8947

Epoch 5/8

15/15 [=====] - 1s 51ms/step - loss: 58.5530 - accuracy: 0.6517 - val_loss: 6.0541 - val_accuracy: 0.9474

Epoch 6/8

15/15 [=====] - 1s 52ms/step - loss: 49.0344 - accuracy: 0.7104 - val_loss: 2.9651 - val_accuracy: 0.9474

Epoch 7/8

15/15 [=====] - 1s 52ms/step - loss: 30.2836 - accuracy: 0.8229 - val_loss: 1.8618 - val_accuracy: 1.0000

Epoch 8/8

15/15 [=====] - 1s 51ms/step - loss: 24.1356 - accuracy: 0.8706 - val_loss: 6.9674 -

val_accuracy: 0.9474

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